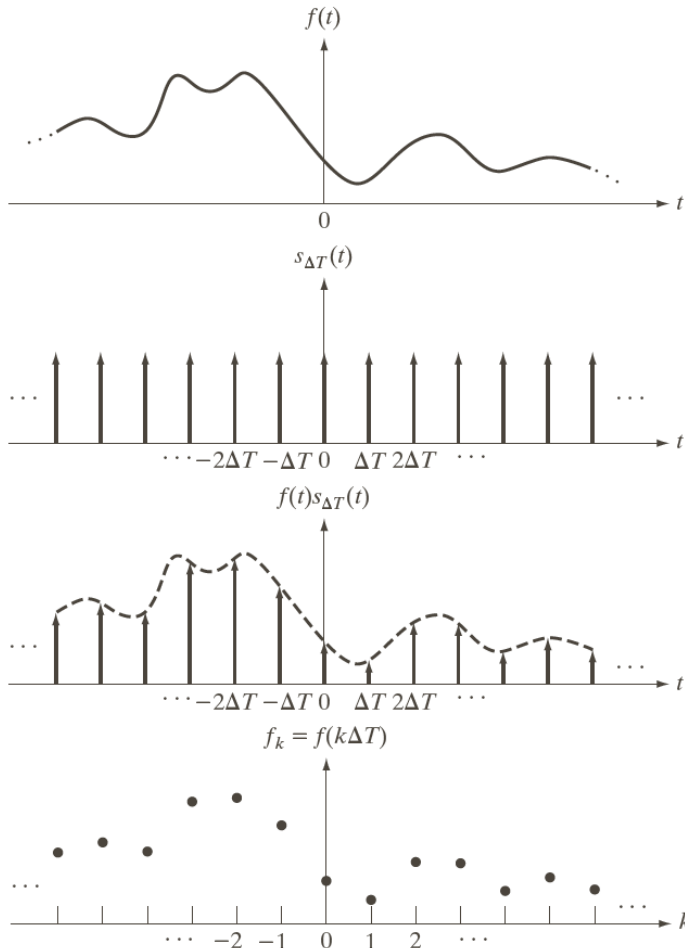


## **Today's Agenda**

---

- **Discrete Time Fourier Transform**
- **Discrete Fourier Transform**
- **2D Fourier Transform**
- **Filtering in Frequency domain**

# Discrete-Time Fourier Transform



**FIGURE 4.5**  
 (a) A continuous function. (b) Train of impulses used to model the sampling process. (c) Sampled function formed as the product of (a) and (b). (d) Sample values obtained by integration and using the sifting property of the impulse. (The dashed line in (c) is shown for reference. It is not part of the data.)

$$\tilde{f}(t) = f(t)s_{\Delta T}(t) = \sum_{k=-\infty}^{\infty} f(t)\delta(t - k\Delta T)$$

$$\tilde{f}(t) = \sum_{k=-\infty}^{\infty} f(k\Delta T)\delta(t - k\Delta T)$$

Discrete data  $\rightarrow$  interval has discrete-time

# Discrete-Time Fourier Transform

---

$$\tilde{f}(t) = f(t)s_{\Delta T}(t) = \sum_{k=-\infty}^{\infty} f(t)\delta(t - k\Delta T)$$

Definition of FT



$$\begin{aligned}\tilde{F}(\mu) &= \int_{-\infty}^{\infty} \tilde{f}(t) e^{-j2\pi\mu t} dt \\ &= \sum_{k=-\infty}^{\infty} f(k\Delta T) e^{-j2\pi\mu k\Delta T}\end{aligned}$$

# Discrete-Time Fourier Transform

---

$$\tilde{f}(t) = f(t)s_{\Delta T}(t) = \sum_{k=-\infty}^{\infty} f(t)\delta(t - k\Delta T)$$



Discrete-Time Fourier Transform

$$\tilde{F}(\mu) = \frac{1}{\Delta T} \sum_{k=-\infty}^{\infty} F\left(\mu - \frac{k}{\Delta T}\right) = \sum_{k=-\infty}^{\infty} f(k\Delta T) e^{-j2\pi\mu k\Delta T}$$

$\tilde{F}(\mu)$  is continuous  $\rightarrow$  Difficult to implement in DSP applications

Sample  $\tilde{F}(\mu)$  in one period  
with M equally spaced  
samples



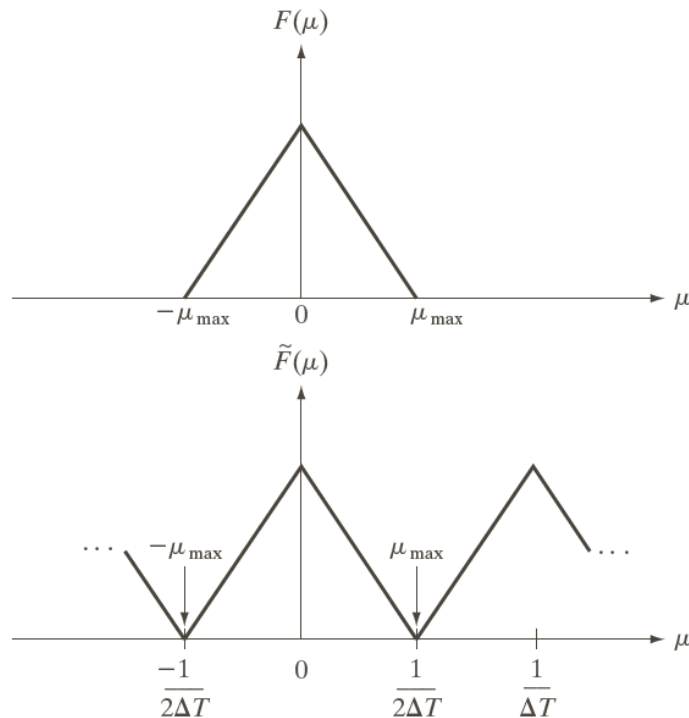
Discrete Fourier  
Transform

## Note that

---

Total  $M$  units in the frequency domain:  $1/\Delta T$

1 unit in frequency domain:  $\Delta\mu = 1/(M\Delta T)$

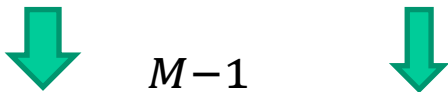


# Discrete Fourier Transform (DFT)

---

coefficients of a  
combination of sinusoids

a finite sequence of  
equally spaced samples


$$F(u) = \sum_{x=0}^{M-1} f(x) e^{-j2\pi xu/M}$$
$$f(x) = \frac{1}{M} \sum_{u=0}^{M-1} F(u) e^{j2\pi ux/M}$$
$$x, u = 0, 1, 2, \dots, M-1$$

- $1/M$  is the sampling interval
- $u$  is an integer  $\rightarrow$  the frequency is an integer multiplier of  $\frac{2\pi}{M}$
- Both input & output are finite

## Discrete FT (DFT)

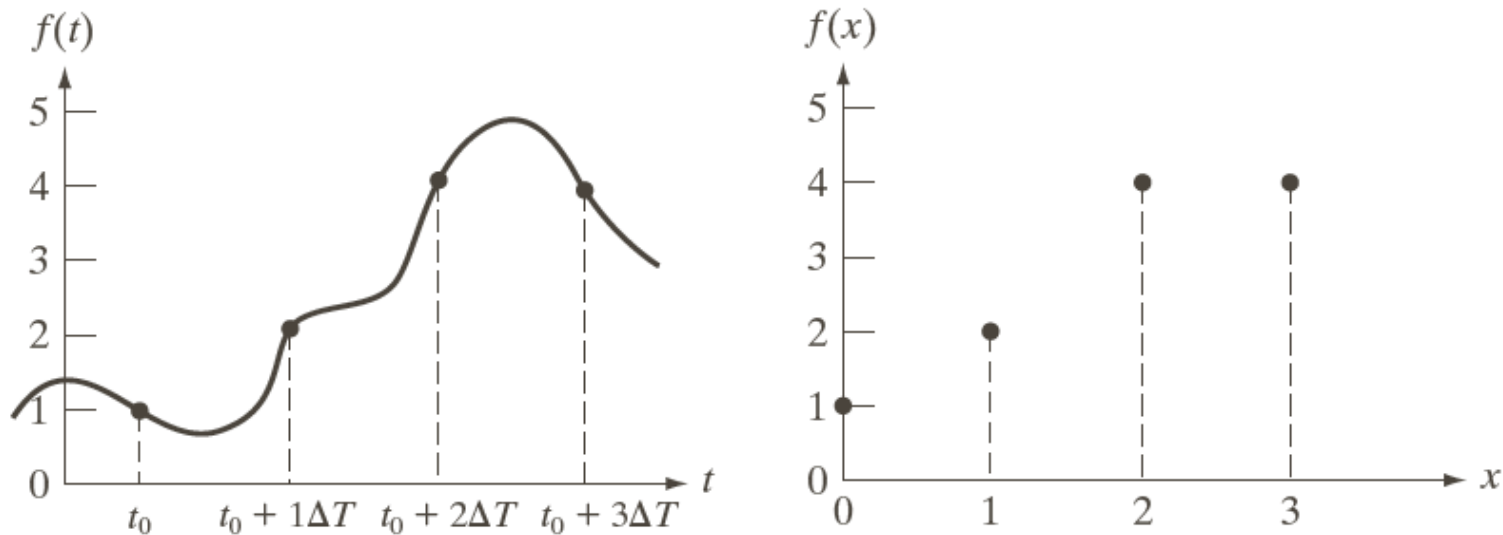
---

### Circular Convolution:

$$f(x) * h(x) = \sum_{m=0}^{M-1} f(m)h(x-m)$$

- The convolution is periodic

## DFT of this Function:



a b

**FIGURE 4.11**  
(a) A function, and (b) samples in the  $x$ -domain. In (a),  $t$  is a continuous variable; in (b),  $x$  represents integer values.

$$F(0) = \sum_{x=0}^3 f(x) = 1 + 2 + 4 + 4 = 11$$

$$F(1) = ?, F(2) = ?, F(3) = ?$$



# FT, FS, DTFT, DFT

---

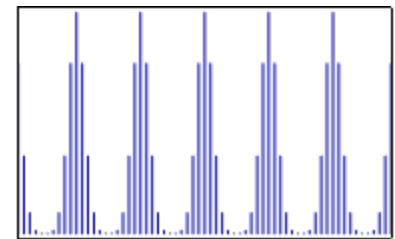
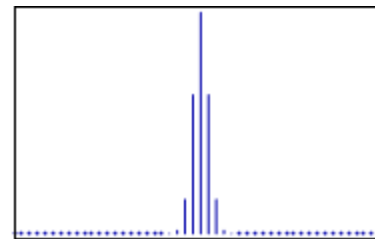
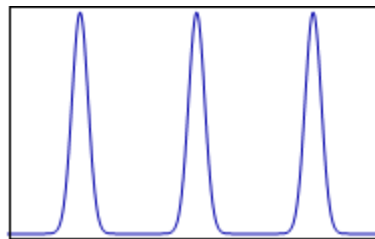
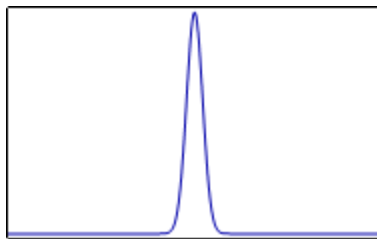
FT

FS

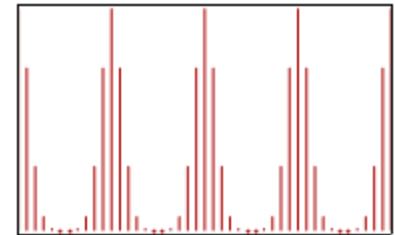
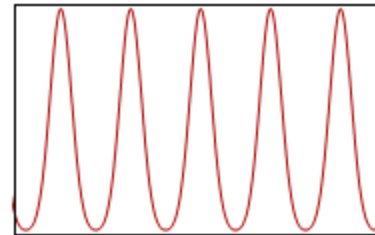
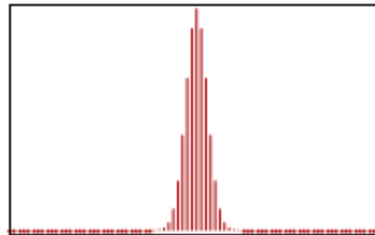
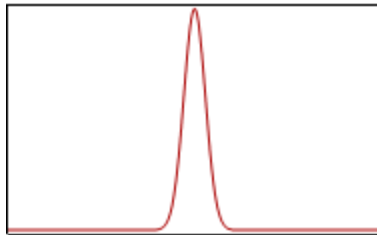
DTFT

DFT

Spatial



Frequency



[https://en.wikipedia.org/wiki/Discrete\\_Fourier\\_transform#/media/File:From\\_Continuous\\_To\\_Discrete\\_Fourier\\_Transform.gif](https://en.wikipedia.org/wiki/Discrete_Fourier_transform#/media/File:From_Continuous_To_Discrete_Fourier_Transform.gif)

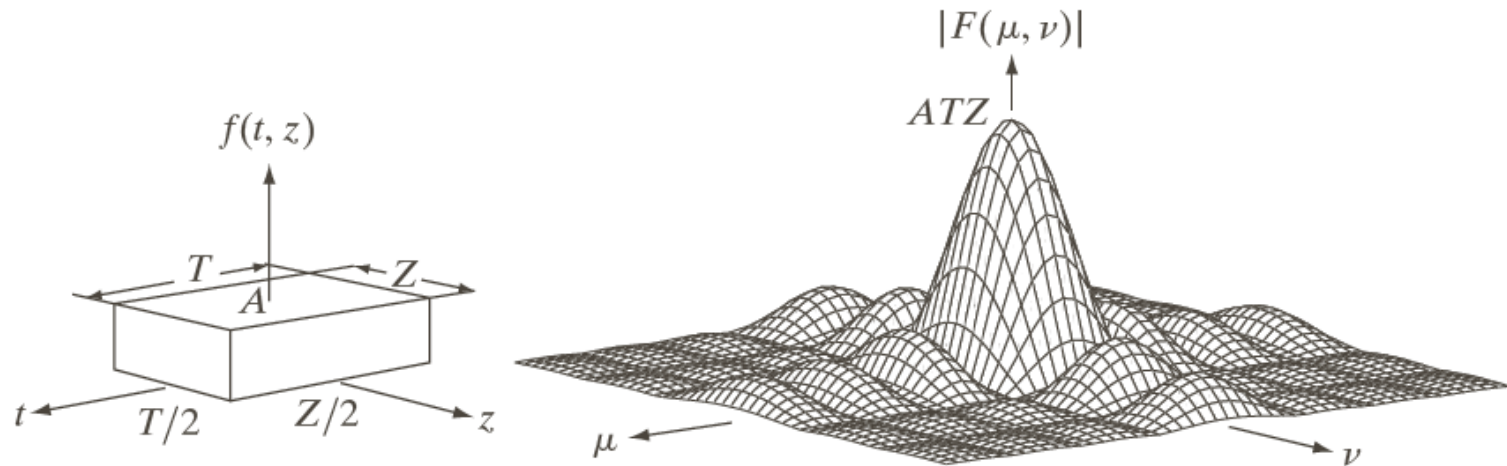
## Extension to 2D Fourier Transform

---

$$F(\mu, \nu) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(t, z) e^{-j2\pi(\mu t + \nu z)} dt dz$$

$$f(t, z) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} F(\mu, \nu) e^{j2\pi(\mu t + \nu z)} d\mu d\nu$$

## 2D Step Function and FT



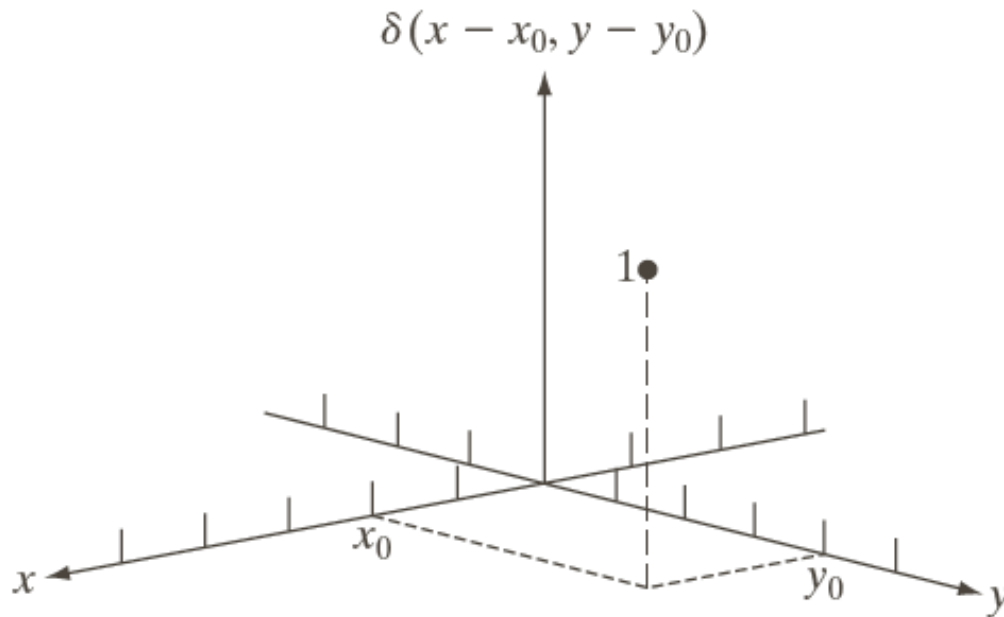
a b

**FIGURE 4.13** (a) A 2-D function, and (b) a section of its spectrum (not to scale). The block is longer along the  $t$ -axis, so the spectrum is more “contracted” along the  $\mu$ -axis. Compare with Fig. 4.4.

$$F(\mu, \nu) = ATZ \left[ \frac{\sin(\pi\mu T)}{(\pi\mu T)} \right] \left[ \frac{\sin(\pi\nu Z)}{(\pi\nu Z)} \right]$$

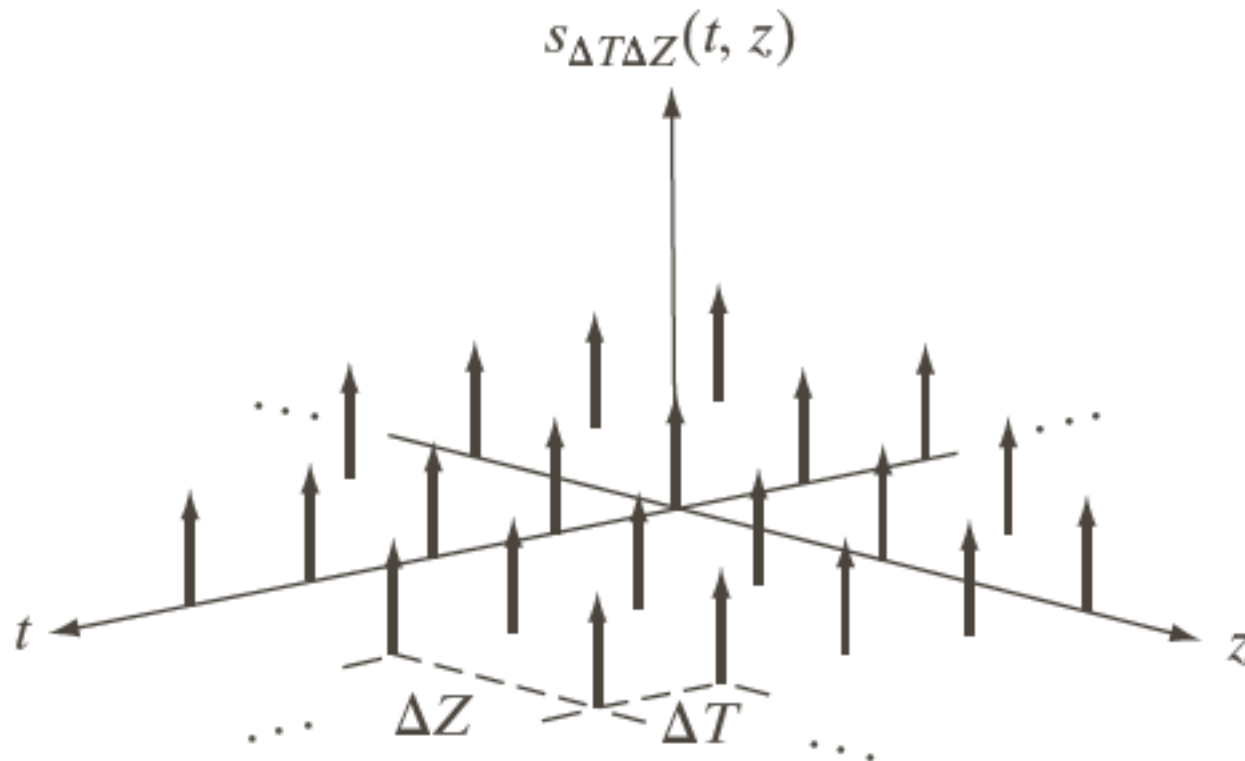
# Extension to 2D Fourier Transform

---



**FIGURE 4.12**  
Two-dimensional unit discrete impulse. Variables  $x$  and  $y$  are discrete, and  $\delta$  is zero everywhere except at coordinates  $(x_0, y_0)$ .

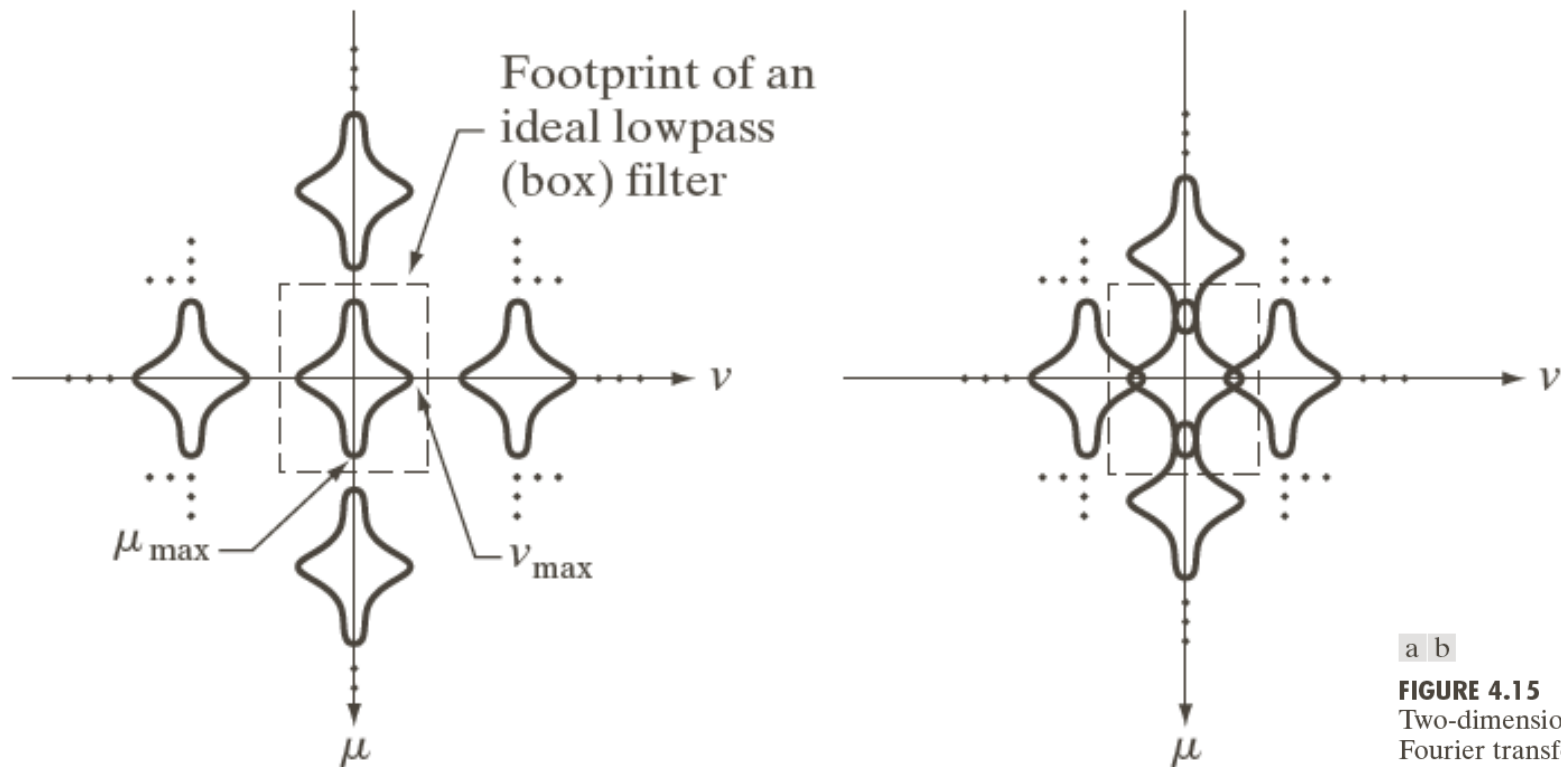
## 2D Sampling



**FIGURE 4.14**  
Two-dimensional  
impulse train.

$$S_{\Delta T \Delta Z}(t, z) = \sum_{m=-\infty}^{\infty} \sum_{n=-\infty}^{\infty} \delta(t - m\Delta T, z - n\Delta Z)$$

# Aliasing in 2D

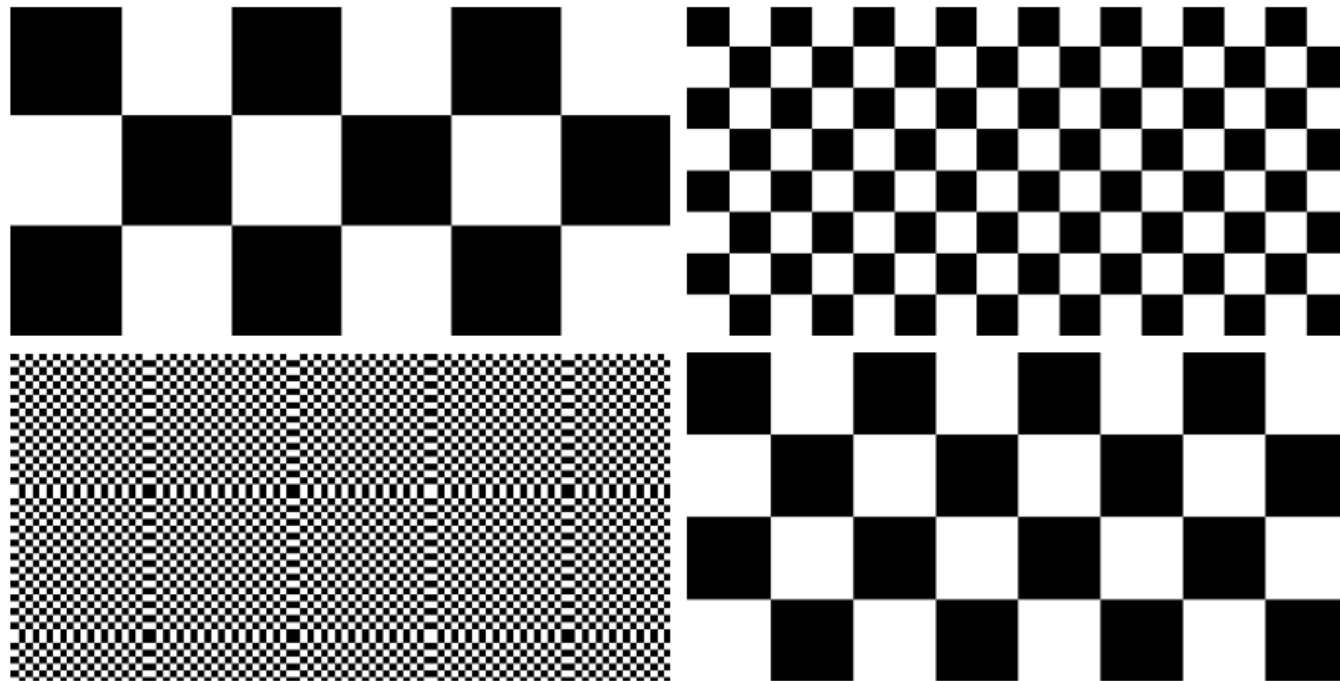


a b

**FIGURE 4.15**  
Two-dimensional  
Fourier transforms  
of (a) an over-  
sampled, and  
(b) under-sampled  
band-limited  
function.

# Aliasing Examples

---



|   |   |
|---|---|
| a | b |
| c | d |

**FIGURE 4.16** Aliasing in images. In (a) and (b), the lengths of the sides of the squares are 16 and 6 pixels, respectively, and aliasing is visually negligible. In (c) and (d), the sides of the squares are 0.9174 and 0.4798 pixels, respectively, and the results show significant aliasing. Note that (d) masquerades as a “normal” image.

# Aliasing Example



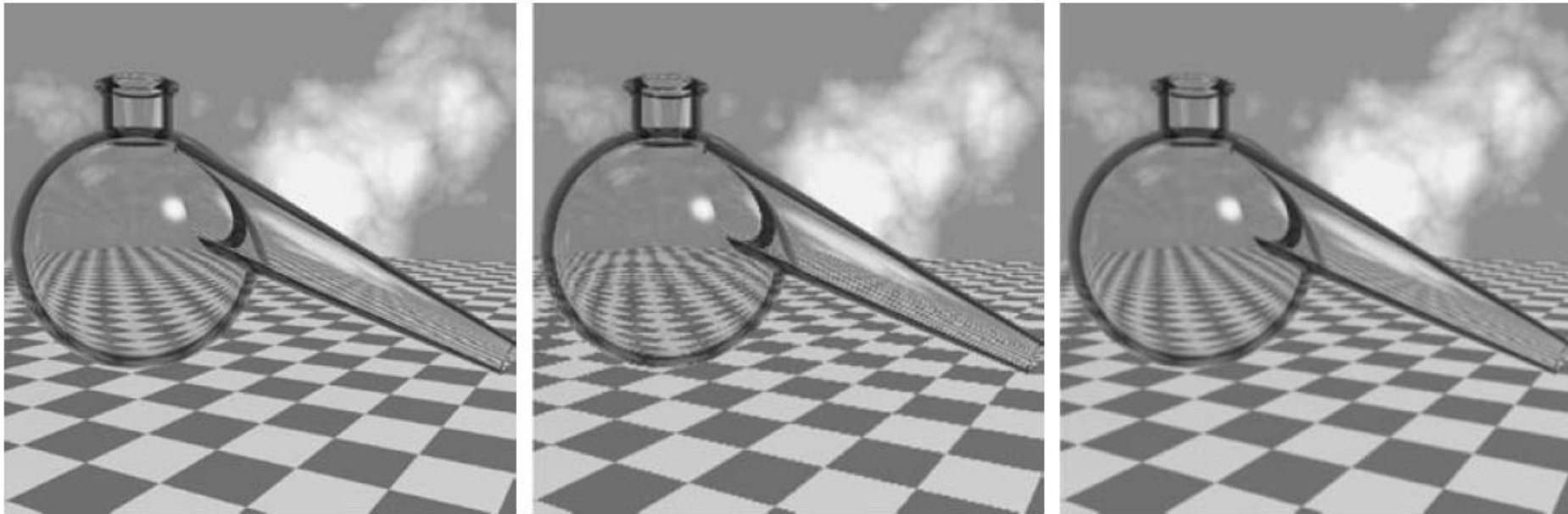
a b c

**FIGURE 4.17** Illustration of aliasing on resampled images. (a) A digital image with negligible visual aliasing. (b) Result of resizing the image to 50% of its original size by pixel deletion. Aliasing is clearly visible. (c) Result of blurring the image in (a) with a  $3 \times 3$  averaging filter prior to resizing. The image is slightly more blurred than (b), but aliasing is not longer objectionable. (Original image courtesy of the Signal Compression Laboratory, University of California, Santa Barbara.)



# Aliasing Example

---



a b c

**FIGURE 4.18** Illustration of jaggies. (a) A  $1024 \times 1024$  digital image of a computer-generated scene with negligible visible aliasing. (b) Result of reducing (a) to 25% of its original size using bilinear interpolation. (c) Result of blurring the image in (a) with a  $5 \times 5$  averaging filter prior to resizing it to 25% using bilinear interpolation. (Original image courtesy of D. P. Mitchell, Mental Landscape, LLC.)

## 2D Discrete Fourier Transform

---

$$F(u, v) = \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} f(x, y) e^{-j2\pi(\frac{ux}{M} + \frac{vy}{N})}$$

$$f(x, y) = \frac{1}{MN} \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} F(u, v) e^{j2\pi(\frac{ux}{M} + \frac{vy}{N})}$$

$$u, x = 0, 1, \dots, M$$

$$v, y = 0, 1, \dots, N$$

Periodicity:

$$F(u, v) = F(u + k_1 M, v + k_2 N)$$
$$f(x, y) = f(x + k_1 M, y + k_2 N)$$

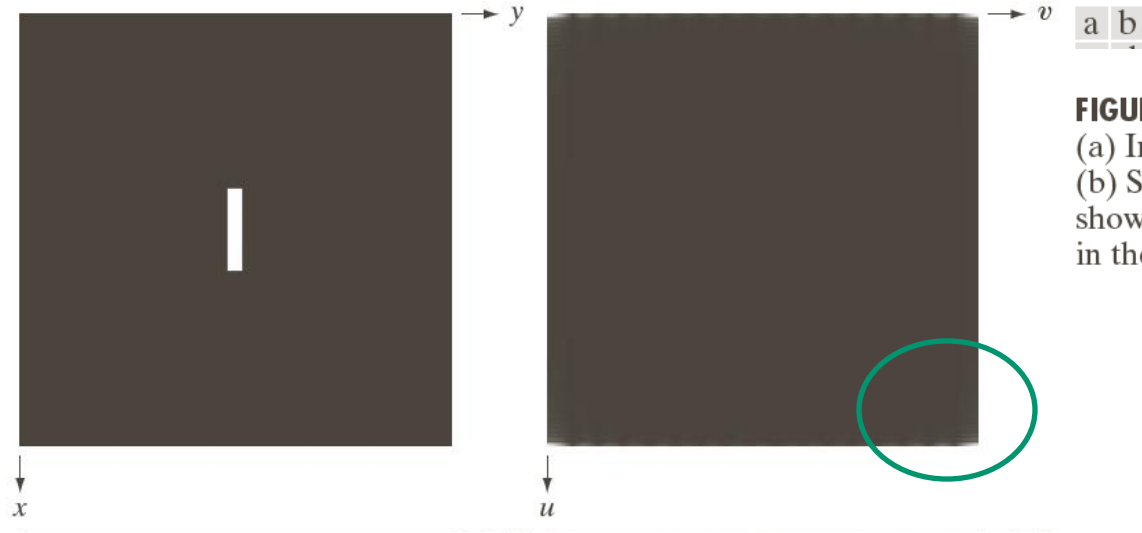
# Fourier Spectrum and Phase Angle

$$F(u, v) = |F(u, v)| e^{j\phi(u, v)}$$

spectrum

$$\phi(u, v) = \arctan \frac{I(u, v)}{R(u, v)}$$

phase



**FIGURE 4.24**  
 (a) Image.  
 (b) Spectrum showing bright spots in the four corners.

$$F(0,0) = MN \bar{f}(x, y)$$

a  
b  
c d

# Centering the DFT

## Modulation/Translation:

$$f(x)e^{j2\pi(u_0x/M)}$$

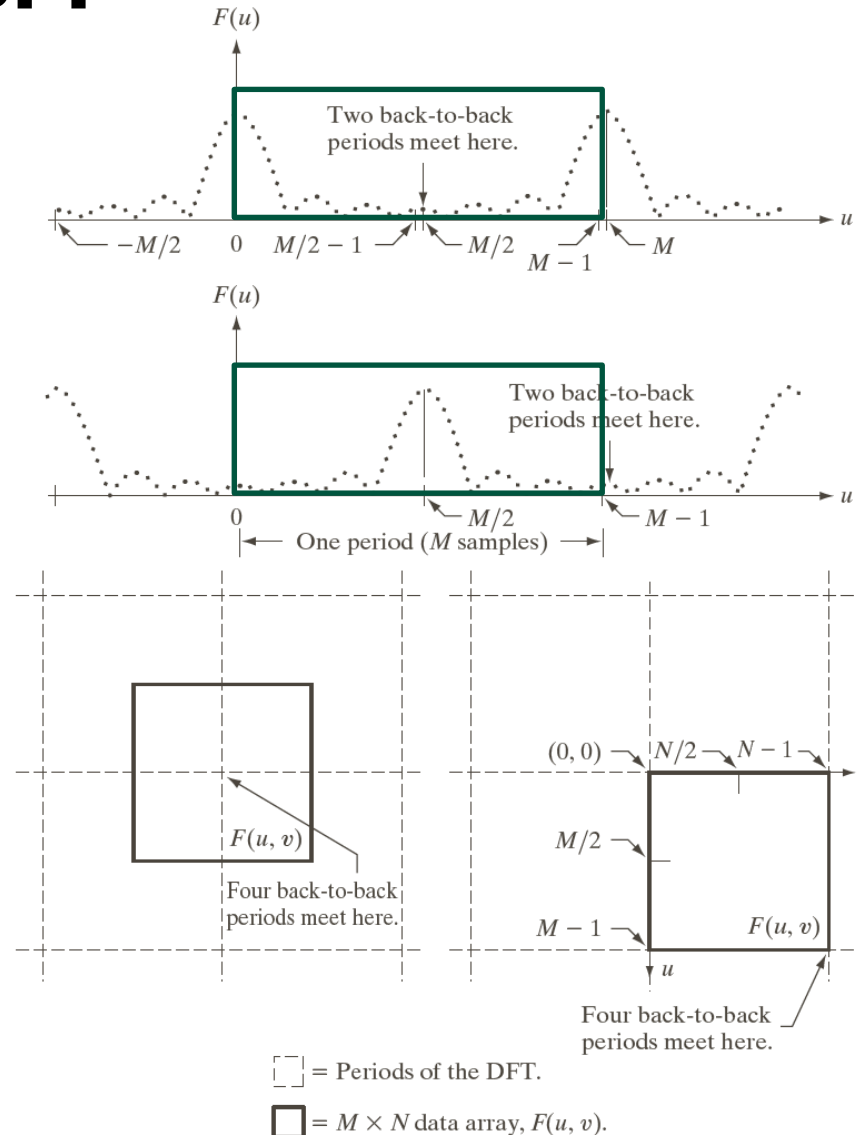
$$\leftrightarrow F(u - u_0)$$

$$f(x)(-1)^x$$

$$\leftrightarrow F(u - M/2)$$

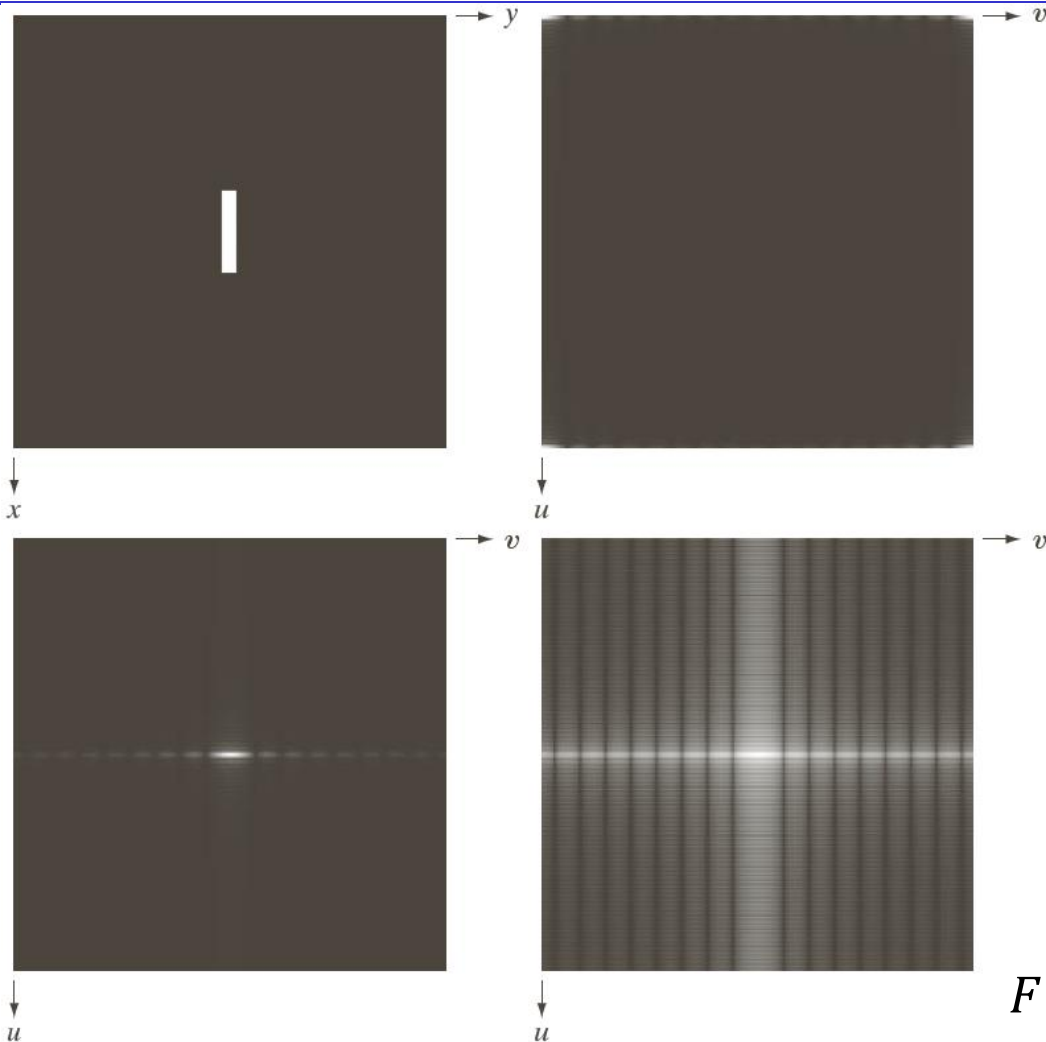
$$f(x, y)(-1)^{x+y}$$

$$\leftrightarrow F(u - M/2, v - N/2)$$



**FIGURE 4.23** Centering the Fourier transform. (a) A 1-D DFT showing an infinite number of periods. (b) Shifted DFT obtained by multiplying  $f(x)$  by  $(-1)^x$  before computing  $F(u)$ . (c) A 2-D DFT showing an infinite number of periods. The solid area is the  $M \times N$  data array,  $F(u, v)$ , obtained with Eq. (4.5-15). This array consists of four quarter periods. (d) A Shifted DFT obtained by multiplying  $f(x, y)$  by  $(-1)^{x+y}$  before computing  $F(u, v)$ . The data now contains one complete, centered period, as in (b).

# Fourier Spectrum and Phase Angle



|   |   |
|---|---|
| a | b |
| c | d |

**FIGURE 4.24**

(a) Image.  
 (b) Spectrum showing bright spots in the four corners.  
 (c) Centered spectrum. (d) Result showing increased detail after a log transformation. The zero crossings of the spectrum are closer in the vertical direction because the rectangle in (a) is longer in that direction. The coordinate convention used throughout the book places the origin of the spatial and frequency domains at the top left.

$$F(\mu, \nu) = ATZ \left[ \frac{\sin(\pi\mu T)}{(\pi\mu T)} \right] \left[ \frac{\sin(\pi\nu Z)}{(\pi\nu Z)} \right]$$

# Fourier Spectrum After Translation and Rotation

|   |   |
|---|---|
| a | b |
| c | d |

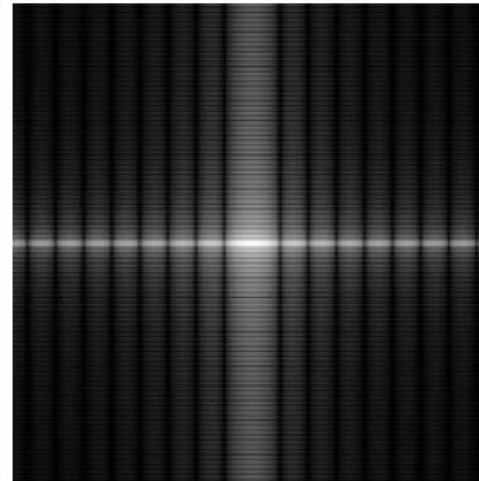
**FIGURE 4.25**

(a) The rectangle in Fig. 4.24(a) translated, and (b) the corresponding spectrum. (c) Rotated rectangle, and (d) the corresponding spectrum. The spectrum corresponding to the translated rectangle is identical to the spectrum corresponding to the original image in Fig. 4.24(a).

$$f(x - x_0, y - y_0) \leftrightarrow F(u, v)e^{-j2\pi(\frac{ux_0}{M} + \frac{vy_0}{N})}$$

$$\Downarrow$$

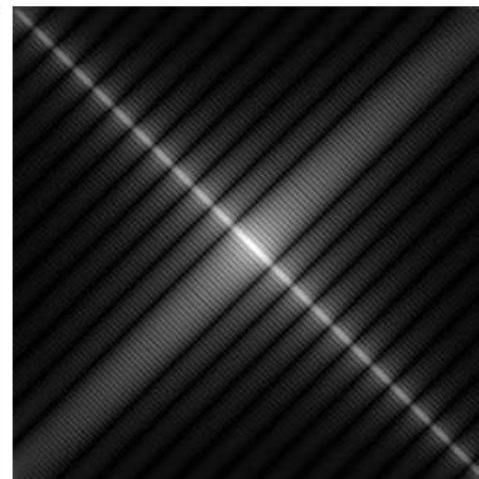
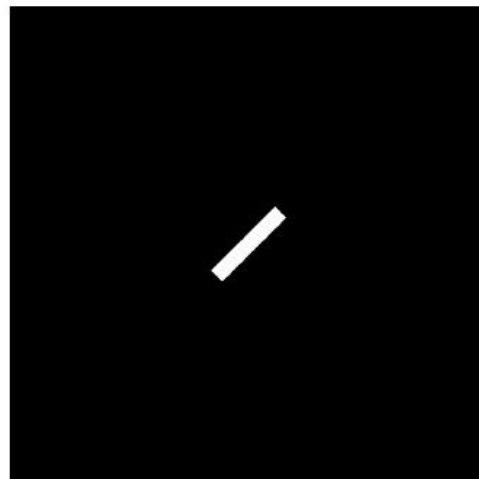
$$|F(u, v)|$$



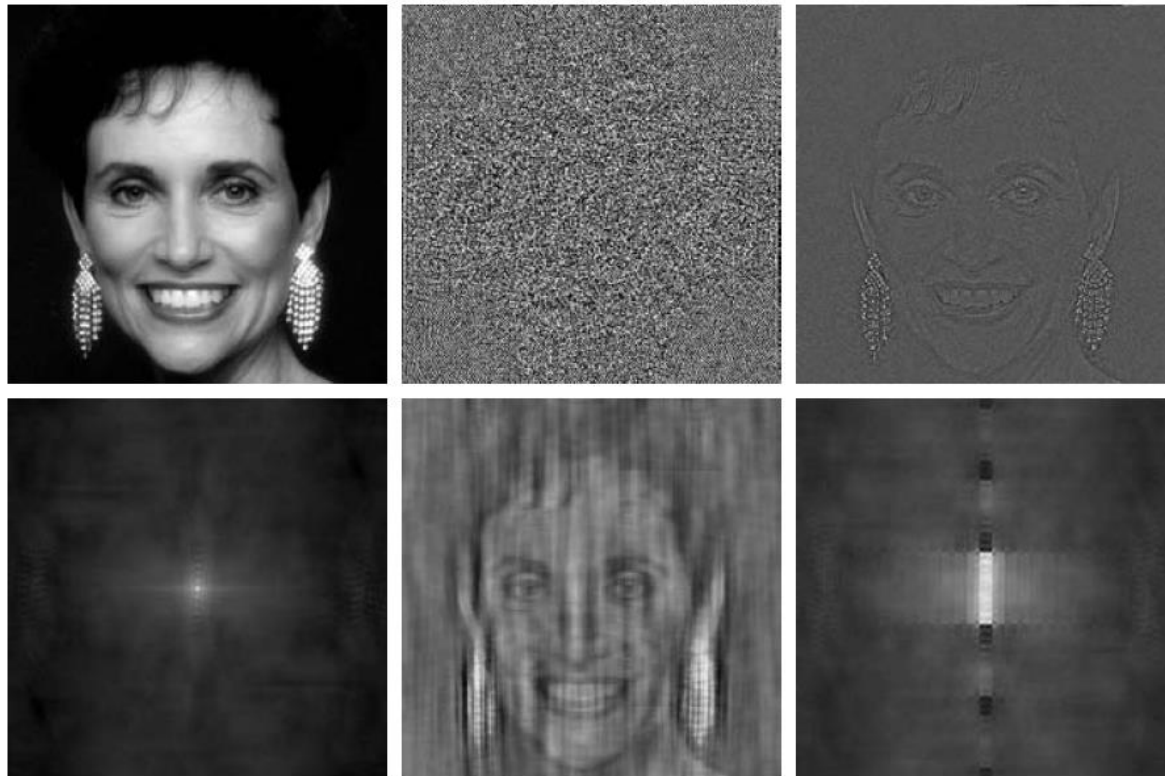
$$f(r, \theta + \theta_0) \leftrightarrow F(w, \varphi + \theta_0)$$

$$x = r \cos \theta, y = r \sin \theta$$

$$u = w \cos \varphi, v = w \sin \varphi$$



# The Roles of Fourier Spectrum and Phase Angle



|   |   |   |
|---|---|---|
| a | b | c |
| d | e | f |

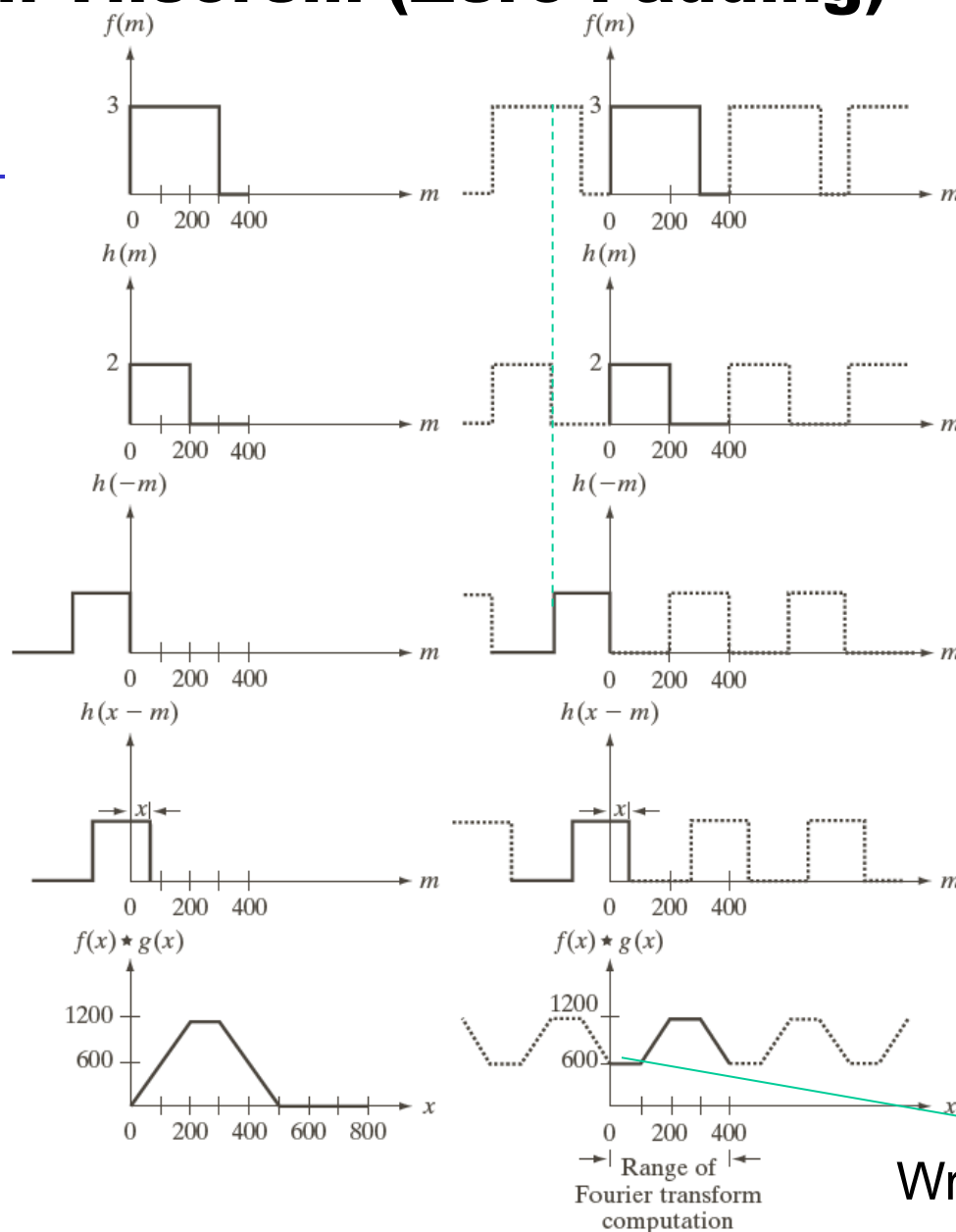
**FIGURE 4.27** (a) Woman. (b) Phase angle. (c) Woman reconstructed using only the phase angle. (d) Woman reconstructed using only the spectrum. (e) Reconstruction using the phase angle corresponding to the woman and the spectrum corresponding to the rectangle in Fig. 4.24(a). (f) Reconstruction using the phase of the rectangle and the spectrum of the woman.



# 2D Convolution Theorem (Zero Padding)

Length for the input signals after zero-padding

$$P \geq A + B - 1$$



|   |   |
|---|---|
| a | f |
| b | g |
| c | h |
| d | i |
| e | j |

**FIGURE 4.28** Left column: convolution of two discrete functions obtained using the approach discussed in Section 3.4.2. The result in (e) is correct. Right column: Convolution of the same functions, but taking into account the periodicity implied by the DFT. Note in (j) how data from adjacent periods produce wraparound error, yielding an incorrect convolution result. To obtain the correct result, function padding must be used.

Wraparound error



# 2D DFT Properties

| Name                                                      | Expression(s)                                                                                   |
|-----------------------------------------------------------|-------------------------------------------------------------------------------------------------|
| 1) Discrete Fourier transform (DFT) of $f(x, y)$          | $F(u, v) = \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} f(x, y) e^{-j2\pi(ux/M + vy/N)}$                   |
| 2) Inverse discrete Fourier transform (IDFT) of $F(u, v)$ | $f(x, y) = \frac{1}{MN} \sum_{u=0}^{M-1} \sum_{v=0}^{N-1} F(u, v) e^{j2\pi(ux/M + vy/N)}$       |
| 3) Polar representation                                   | $F(u, v) =  F(u, v)  e^{j\phi(u, v)}$                                                           |
| 4) Spectrum                                               | $ F(u, v)  = [R^2(u, v) + I^2(u, v)]^{1/2}$ $R = \text{Real}(F); \quad I = \text{Imag}(F)$      |
| 5) Phase angle                                            | $\phi(u, v) = \tan^{-1} \left[ \frac{I(u, v)}{R(u, v)} \right]$                                 |
| 6) Power spectrum                                         | $P(u, v) =  F(u, v) ^2$                                                                         |
| 7) Average value                                          | $\bar{f}(x, y) = \frac{1}{MN} \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} f(x, y) = \frac{1}{MN} F(0, 0)$ |

**TABLE 4.2**  
Summary of DFT definitions and corresponding expressions.

(Continued)

## 2D DFT Properties (cont.)

| Name                                                                             | Expression(s)                                                                                                                                                                                                                                                                                                                                                                           |
|----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 8) Periodicity ( $k_1$ and $k_2$ are integers)                                   | $F(u, v) = F(u + k_1M, v) = F(u, v + k_2N)$ $= F(u + k_1M, v + k_2N)$ $f(x, y) = f(x + k_1M, y) = f(x, y + k_2N)$ $= f(x + k_1M, y + k_2N)$                                                                                                                                                                                                                                             |
| 9) Convolution                                                                   | $f(x, y) \star h(x, y) = \sum_{m=0}^{M-1} \sum_{n=0}^{N-1} f(m, n)h(x - m, y - n)$                                                                                                                                                                                                                                                                                                      |
| 10) Correlation                                                                  | $f(x, y) \star h(x, y) = \sum_{m=0}^{M-1} \sum_{n=0}^{N-1} f^*(m, n)h(x + m, y + n)$                                                                                                                                                                                                                                                                                                    |
| 11) Separability                                                                 | The 2-D DFT can be computed by computing 1-D DFT transforms along the rows (columns) of the image, followed by 1-D transforms along the columns (rows) of the result. See Section 4.11.1.                                                                                                                                                                                               |
| 12) Obtaining the inverse Fourier transform using a forward transform algorithm. | $MNf^*(x, y) = \sum_{u=0}^{M-1} \sum_{v=0}^{N-1} F^*(u, v)e^{-j2\pi(ux/M+vy/N)}$ <p>This equation indicates that inputting <math>F^*(u, v)</math> into an algorithm that computes the forward transform (right side of above equation) yields <math>MNf^*(x, y)</math>. Taking the complex conjugate and dividing by <math>MN</math> gives the desired inverse. See Section 4.11.2.</p> |

TABLE 4.2  
(Continued)

# 2D DFT Properties (Symmetry)

| Spatial Domain <sup>†</sup> |                              | Frequency Domain <sup>†</sup>                 |
|-----------------------------|------------------------------|-----------------------------------------------|
| 1)                          | $f(x, y)$ real               | $\Leftrightarrow F^*(u, v) = F(-u, -v)$       |
| 2)                          | $f(x, y)$ imaginary          | $\Leftrightarrow F^*(-u, -v) = -F(u, v)$      |
| 3)                          | $f(x, y)$ real               | $\Leftrightarrow R(u, v)$ even; $I(u, v)$ odd |
| 4)                          | $f(x, y)$ imaginary          | $\Leftrightarrow R(u, v)$ odd; $I(u, v)$ even |
| 5)                          | $f(-x, -y)$ real             | $\Leftrightarrow F^*(u, v)$ complex           |
| 6)                          | $f(-x, -y)$ complex          | $\Leftrightarrow F(-u, -v)$ complex           |
| 7)                          | $f^*(x, y)$ complex          | $\Leftrightarrow F^*(-u - v)$ complex         |
| 8)                          | $f(x, y)$ real and even      | $\Leftrightarrow F(u, v)$ real and even       |
| 9)                          | $f(x, y)$ real and odd       | $\Leftrightarrow F(u, v)$ imaginary and odd   |
| 10)                         | $f(x, y)$ imaginary and even | $\Leftrightarrow F(u, v)$ imaginary and even  |
| 11)                         | $f(x, y)$ imaginary and odd  | $\Leftrightarrow F(u, v)$ real and odd        |
| 12)                         | $f(x, y)$ complex and even   | $\Leftrightarrow F(u, v)$ complex and even    |
| 13)                         | $f(x, y)$ complex and odd    | $\Leftrightarrow F(u, v)$ complex and odd     |

**TABLE 4.1** Some symmetry properties of the 2-D DFT and its inverse.  $R(u, v)$  and  $I(u, v)$  are the real and imaginary parts of  $F(u, v)$ , respectively. The term *complex* indicates that a function has nonzero real and imaginary parts.

<sup>†</sup>Recall that  $x, y, u$ , and  $v$  are *discrete* (integer) variables, with  $x$  and  $u$  in the range  $[0, M - 1]$ , and  $y$ , and  $v$  in the range  $[0, N - 1]$ . To say that a complex function is *even* means that its real *and* imaginary parts are even, and similarly for an odd complex function.

Even (symmetric):  $w_e(x, y) = w_e(M - x, N - y)$

Odd (antisymmetric):  $w_o(x, y) = -w_o(M - x, N - y)$

# 2D DFT Properties (cont.)

**TABLE 4.3**

Summary of DFT pairs. The closed-form expressions in 12 and 13 are valid only for continuous variables. They can be used with discrete variables by sampling the closed-form, continuous expressions.

| Name                                                              | DFT Pairs                                                                                                                                                                          |
|-------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1) Symmetry properties                                            | See Table 4.1                                                                                                                                                                      |
| 2) Linearity                                                      | $af_1(x, y) + bf_2(x, y) \Leftrightarrow aF_1(u, v) + bF_2(u, v)$                                                                                                                  |
| 3) Translation (general)                                          | $f(x, y)e^{j2\pi(u_0x/M+v_0y/N)} \Leftrightarrow F(u - u_0, v - v_0)$<br>$f(x - x_0, y - y_0) \Leftrightarrow F(u, v)e^{-j2\pi(ux_0/M+vy_0/N)}$                                    |
| 4) Translation to center of the frequency rectangle, $(M/2, N/2)$ | $f(x, y)(-1)^{x+y} \Leftrightarrow F(u - M/2, v - N/2)$<br>$f(x - M/2, y - N/2) \Leftrightarrow F(u, v)(-1)^{u+v}$                                                                 |
| 5) Rotation                                                       | $f(r, \theta + \theta_0) \Leftrightarrow F(\omega, \varphi + \theta_0)$<br>$x = r \cos \theta \quad y = r \sin \theta \quad u = \omega \cos \varphi \quad v = \omega \sin \varphi$ |
| 6) Convolution theorem <sup>†</sup>                               | $f(x, y) \star h(x, y) \Leftrightarrow F(u, v)H(u, v)$<br>$f(x, y)h(x, y) \Leftrightarrow F(u, v) \star H(u, v)$                                                                   |

(Continued)

## 2D DFT Properties (cont.)

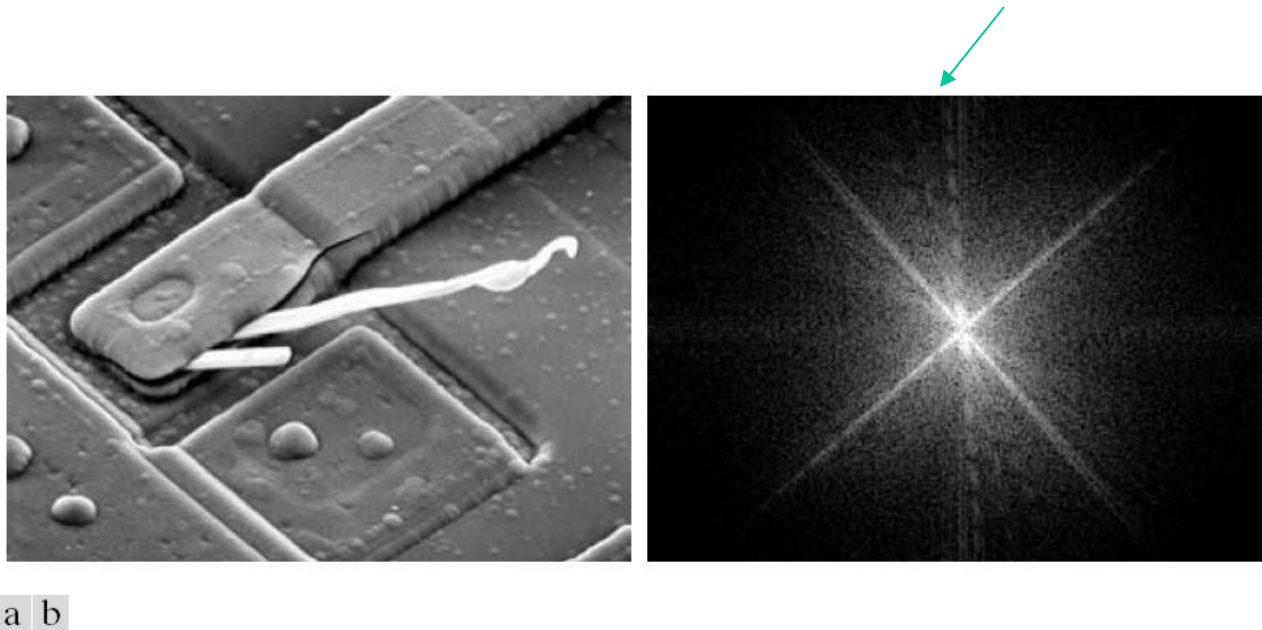
| Name                                                                                                                                                                                                                                                                                                                  | DFT Pairs                                                                                                                                                                                                                                                                                                                     |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7) Correlation theorem <sup>†</sup>                                                                                                                                                                                                                                                                                   | $f(x, y) \star h(x, y) \Leftrightarrow F^*(u, v) H(u, v)$<br>$f^*(x, y) h(x, y) \Leftrightarrow F(u, v) \star H(u, v)$                                                                                                                                                                                                        |
| 8) Discrete unit impulse                                                                                                                                                                                                                                                                                              | $\delta(x, y) \Leftrightarrow 1$                                                                                                                                                                                                                                                                                              |
| 9) Rectangle                                                                                                                                                                                                                                                                                                          | $\text{rect}[a, b] \Leftrightarrow ab \frac{\sin(\pi ua)}{(\pi ua)} \frac{\sin(\pi vb)}{(\pi vb)} e^{-j\pi(ua+vb)}$                                                                                                                                                                                                           |
| 10) Sine                                                                                                                                                                                                                                                                                                              | $\sin(2\pi u_0 x + 2\pi v_0 y) \Leftrightarrow$<br>$j \frac{1}{2} [\delta(u + Mu_0, v + Nv_0) - \delta(u - Mu_0, v - Nv_0)]$                                                                                                                                                                                                  |
| 11) Cosine                                                                                                                                                                                                                                                                                                            | $\cos(2\pi u_0 x + 2\pi v_0 y) \Leftrightarrow$<br>$\frac{1}{2} [\delta(u + Mu_0, v + Nv_0) + \delta(u - Mu_0, v - Nv_0)]$                                                                                                                                                                                                    |
| <p>The following Fourier transform pairs are derivable only for continuous variables, denoted as before by <math>t</math> and <math>z</math> for spatial variables and by <math>\mu</math> and <math>\nu</math> for frequency variables. These results can be used for DFT work by sampling the continuous forms.</p> |                                                                                                                                                                                                                                                                                                                               |
| 12) <i>Differentiation</i><br>(The expressions on the right assume that $f(\pm\infty, \pm\infty) = 0$ .)                                                                                                                                                                                                              | $\left(\frac{\partial}{\partial t}\right)^m \left(\frac{\partial}{\partial z}\right)^n f(t, z) \Leftrightarrow (j2\pi\mu)^m (j2\pi\nu)^n F(\mu, \nu)$<br>$\frac{\partial^m f(t, z)}{\partial t^m} \Leftrightarrow (j2\pi\mu)^m F(\mu, \nu); \frac{\partial^n f(t, z)}{\partial z^n} \Leftrightarrow (j2\pi\nu)^n F(\mu, \nu)$ |
| 13) <i>Gaussian</i>                                                                                                                                                                                                                                                                                                   | $A 2\pi\sigma^2 e^{-2\pi^2\sigma^2(t^2+z^2)} \Leftrightarrow A e^{-(\mu^2+\nu^2)/2\sigma^2}$ ( $A$ is a constant)                                                                                                                                                                                                             |

**TABLE 4.3**  
(Continued)

<sup>†</sup> Assumes that the functions have been extended by zero padding. Convolution and correlation are associative, commutative, and distributive.

# Basics of Filtering in Frequency Domain

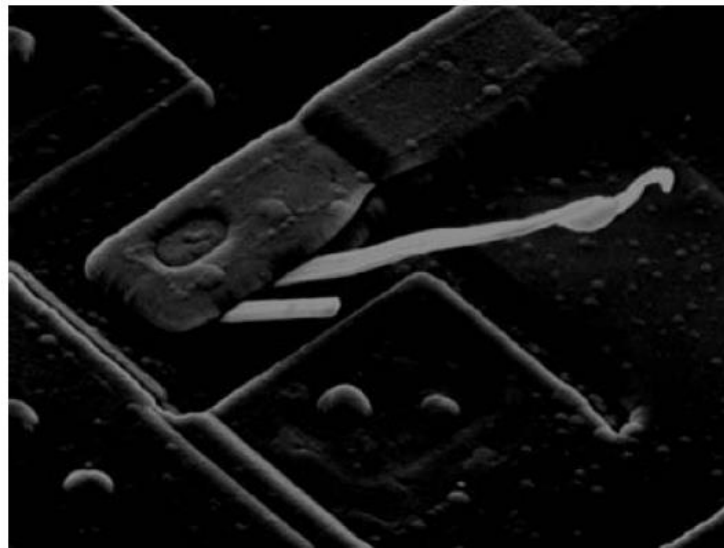
---



**FIGURE 4.29** (a) SEM image of a damaged integrated circuit. (b) Fourier spectrum of (a). (Original image courtesy of Dr. J. M. Hudak, Brockhouse Institute for Materials Research, McMaster University, Hamilton, Ontario, Canada.)

# Frequency Domain Filtering Fundamentals

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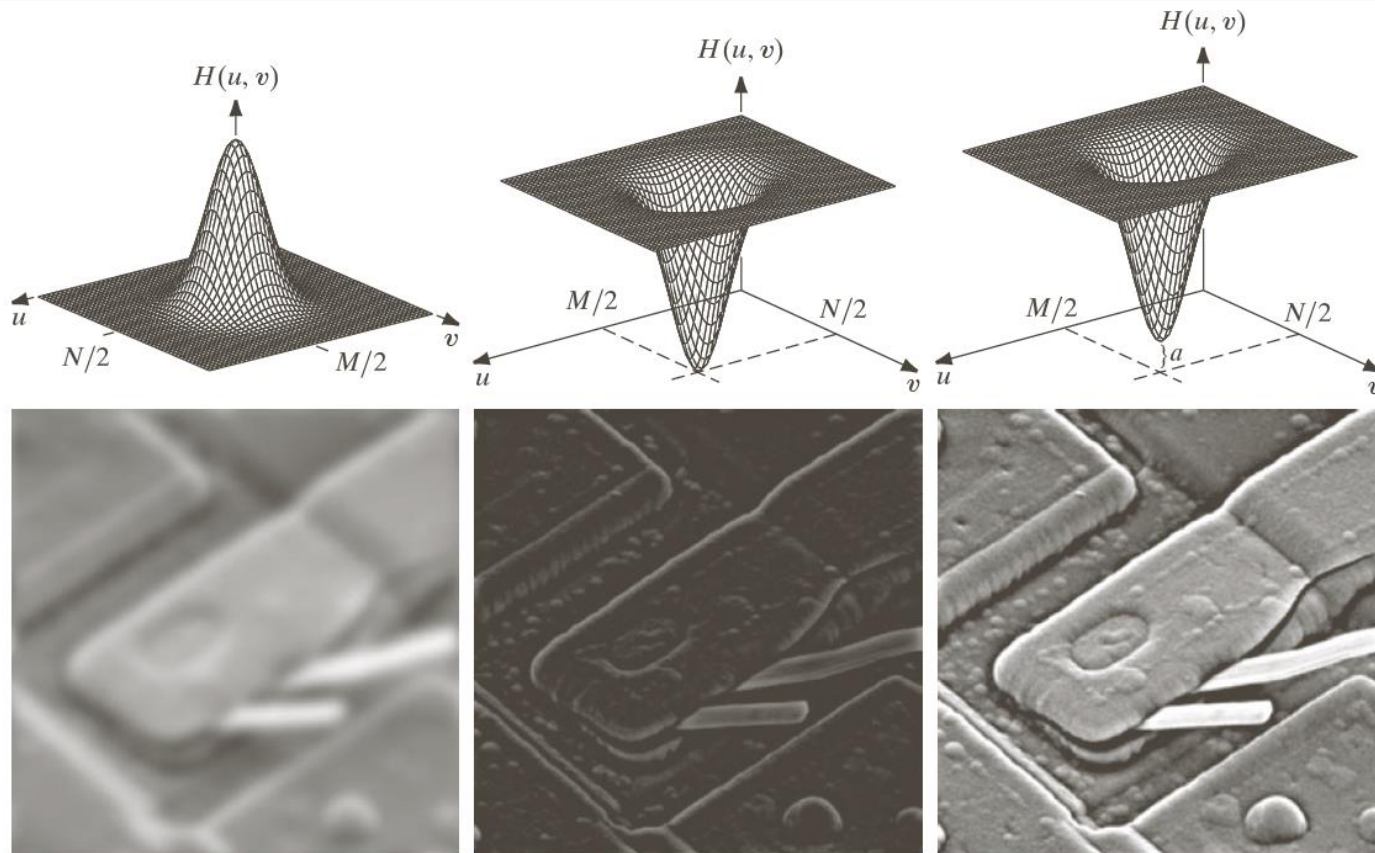
**FIGURE 4.30**  
Result of filtering  
the image in  
Fig. 4.29(a) by  
setting to 0 the  
term  $F(M/2, N/2)$   
in the Fourier  
transform.

$$g(x, y) \leftrightarrow H(u, v)F(u, v)$$

$$\text{Centered } F(M/2, N/2) = MN\bar{f}(x, y) \longrightarrow \text{dc term}$$



# Low-Pass and High-Pass Filters



|   |   |   |
|---|---|---|
| a | b | c |
| d | e | f |

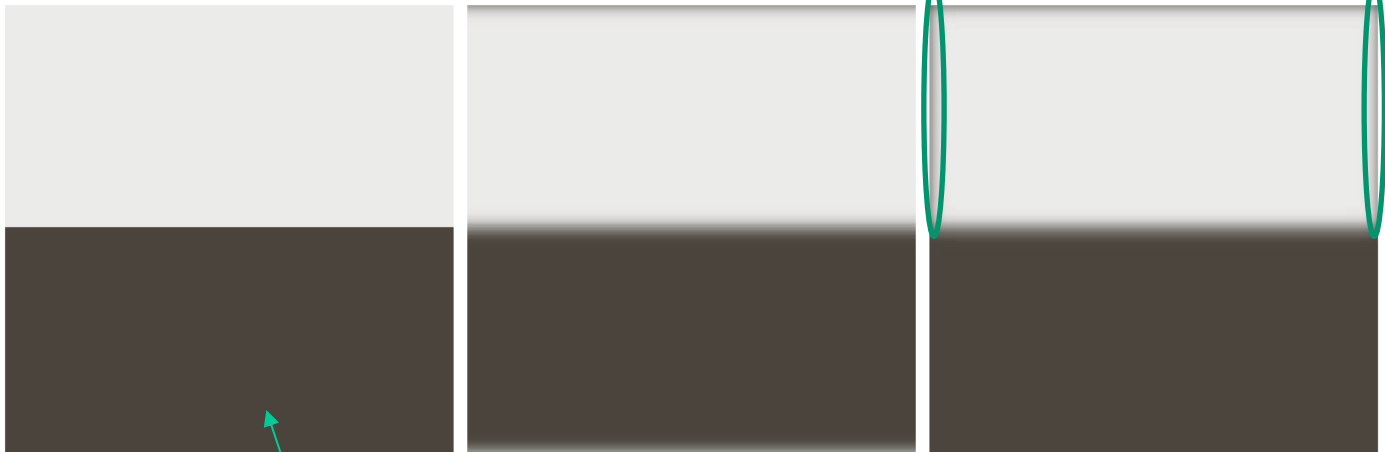
**FIGURE 4.31** Top row: frequency domain filters. Bottom row: corresponding filtered images obtained using Eq. (4.7-1). We used  $a = 0.85$  in (c) to obtain (f) (the height of the filter itself is 1). Compare (f) with Fig. 4.29(a).



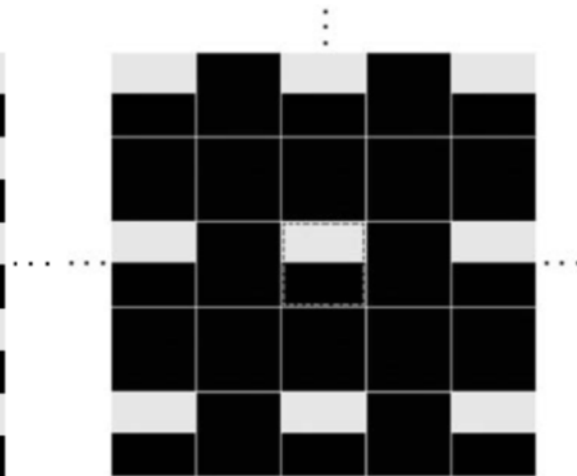
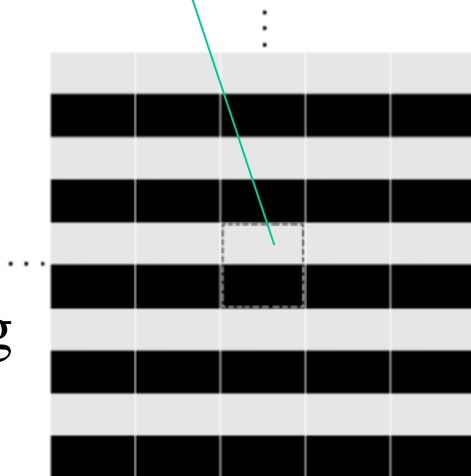
# Effects of Zero-Padding

Smoothing, no padding

Smoothing, padded

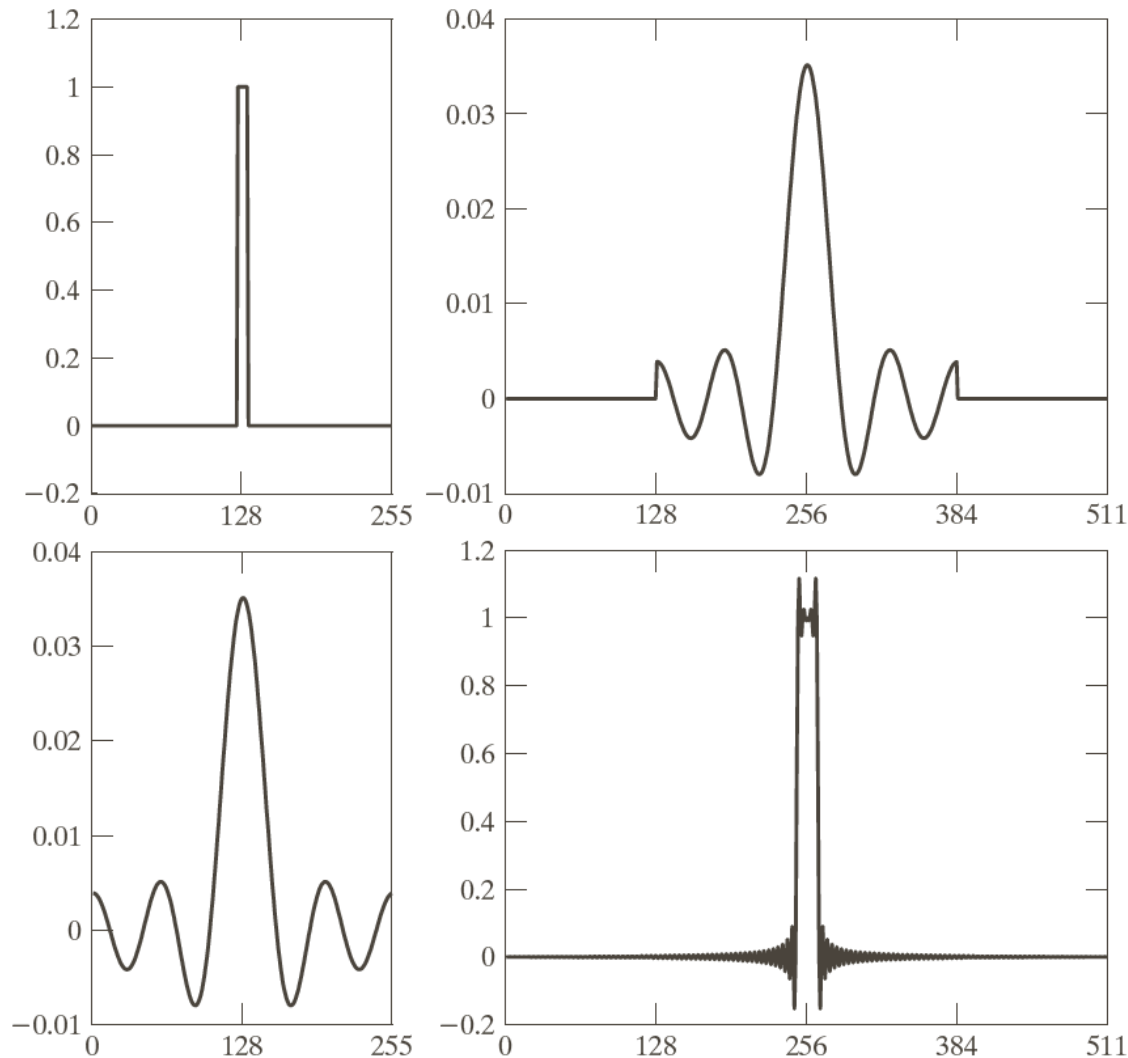


No padding



Padded

# Spatial Zero-Padding for the Filter



**FIGURE 4.34**

(a) Original filter specified in the (centered) frequency domain. (b) Spatial representation obtained by computing the IDFT of (a). (c) Result of padding (b) to twice its length (note the discontinuities). (d) Corresponding filter in the frequency domain obtained by computing the DFT of (c). Note the ringing caused by the discontinuities in (c). (The curves appear continuous because the points were joined to simplify visual analysis.)

## Summary: Filtering in the Frequency Domain

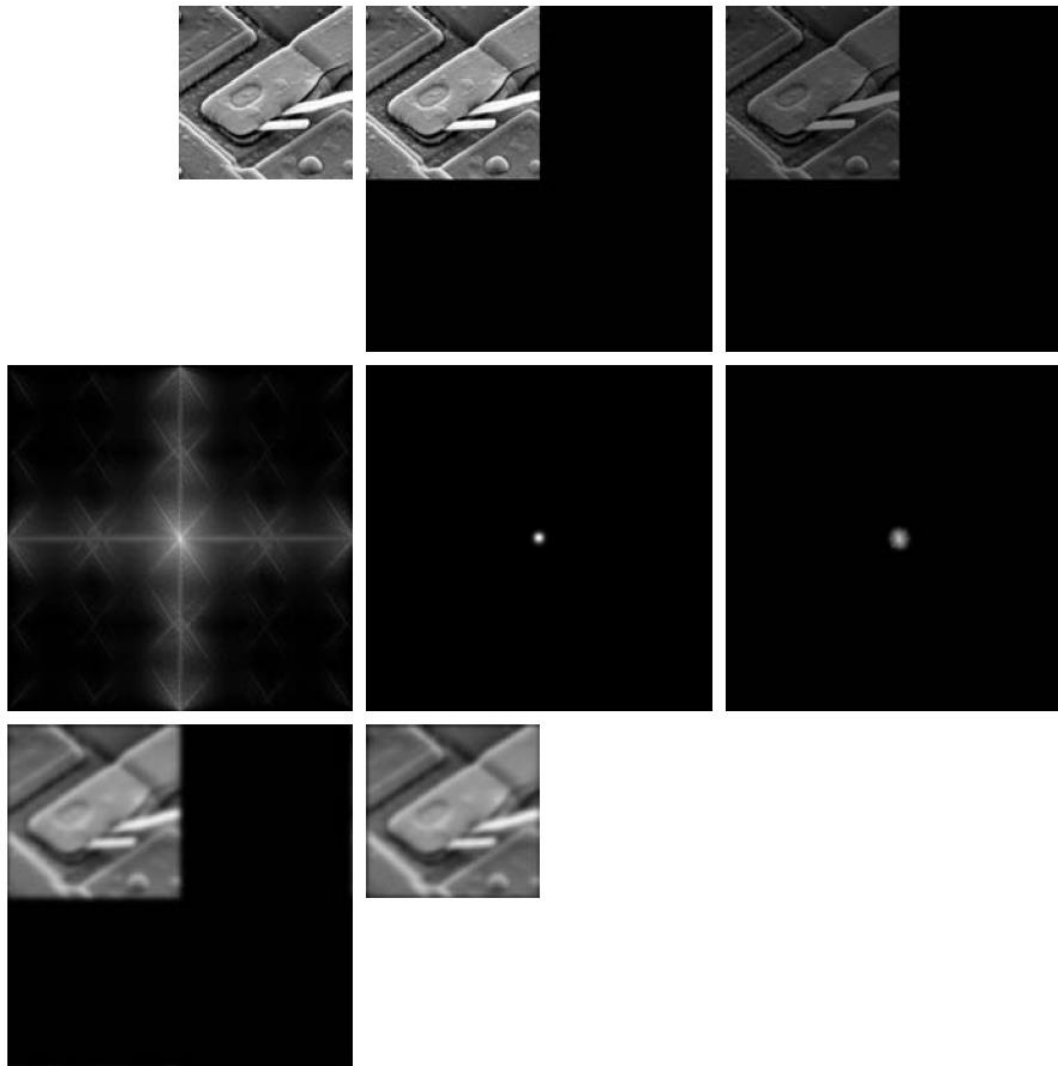
---

1. For  $f(x, y)$ , find  $P = 2M, Q = 2N$
2. Form a padded image  $f_p(x, y)$
3. Centering: Multiply  $f_p(x, y)$  by  $(-1)^{x+y}$
4. Compute DFT  $F(u, v)$
5. Generate a real symmetric filter  $H(u, v)$  of a size  $P \times Q$ , centered at  $(P/2, Q/2)$  and make  $G(u, v) = H(u, v)F(u, v)$
6. Obtain the processed image (should be real in theory)

$$g_p(x, y) = \{\text{real}[g(x, y)]\}(-1)^{x+y}$$

7. Obtained the final processed results  $g(x, y)$  from the top left  $M \times N$  region

# An Example

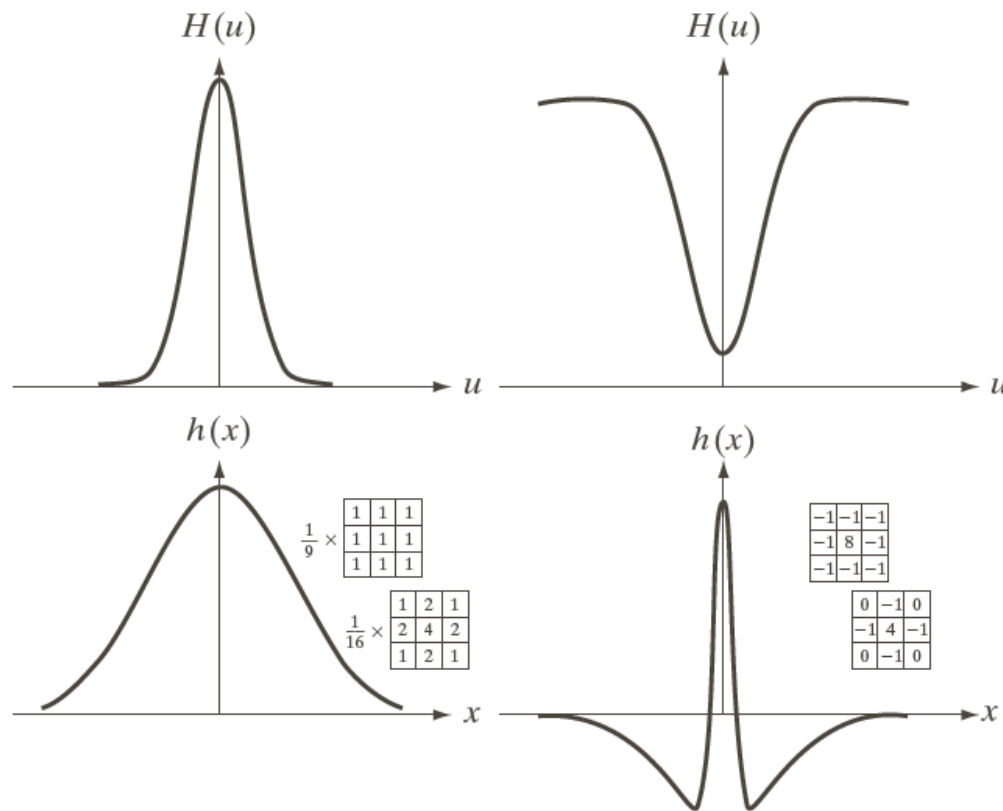


|   |   |   |
|---|---|---|
| a | b | c |
| d | e | f |
| g | h |   |

**FIGURE 4.36**

- (a) An  $M \times N$  image,  $f$ .
- (b) Padded image,  $f_p$  of size  $P \times Q$ .
- (c) Result of multiplying  $f_p$  by  $(-1)^{x+y}$ .
- (d) Spectrum of  $F_p$ .
- (e) Centered Gaussian lowpass filter,  $H$ , of size  $P \times Q$ .
- (f) Spectrum of the product  $HF_p$ .
- (g)  $g_p$ , the product of  $(-1)^{x+y}$  and the real part of the IDFT of  $HF_p$ .
- (h) Final result,  $g$ , obtained by cropping the first  $M$  rows and  $N$  columns of  $g_p$ .

# Correspondence to the Spatial Domain Filter



|   |   |
|---|---|
| a | c |
| b | d |

**FIGURE 4.37**

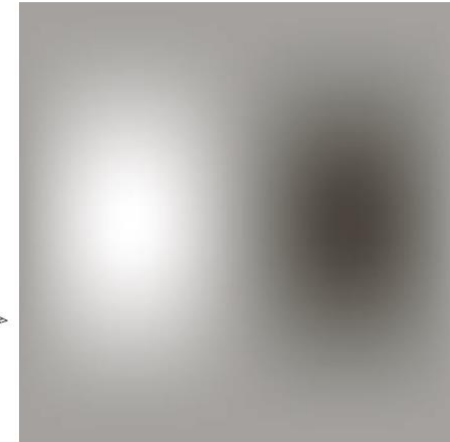
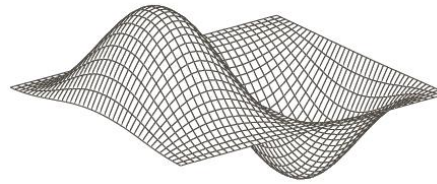
(a) A 1-D Gaussian lowpass filter in the frequency domain. (b) Spatial lowpass filter corresponding to (a). (c) Gaussian highpass filter in the frequency domain. (d) Spatial highpass filter corresponding to (c). The small 2-D masks shown are spatial filters we used in Chapter 3.

**The FT of a Gaussian function is still a Gaussian function**

# An Example (Sobel Mask)



|    |   |   |
|----|---|---|
| -1 | 0 | 1 |
| -2 | 0 | 2 |
| -1 | 0 | 1 |



|   |   |
|---|---|
| a | b |
| c | d |

**FIGURE 4.39**

(a) A spatial mask and perspective plot of its corresponding frequency domain filter. (b) Filter shown as an image. (c) Result of filtering Fig. 4.38(a) in the frequency domain with the filter in (b). (d) Result of filtering the same image with the spatial filter in (a). The results are identical.

